

Decomposition of Concentration-Index Trees

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1 Results

1.1 Exploratory Data Analysis

In the DRC Data Set ([The DHS Program, 2007](#)), 10.32% of children died before reaching age five (Figure 1). Deaths were more commonly observed among children from disadvantaged or rural backgrounds. The children who died differed descriptively from those who survived. Low-wealth households accounted for a larger share of deaths than of survivors (51.3% versus 42.8%). The same pattern was present for rural residence (69.3% versus 60.7%), no skilled attendance at birth (65.6% versus 56.8%), and mothers with no education (67.5% versus 55.2%) (Table 2). Mortality also differed across provinces, ranging from 6.2% in Kinshasa to 14.2% in Katanga (Figure 9). Next we use the direct and indirect methods for decomposition.



Figure 1: Under-five mortality distribution and DHS wealth index in the DRC analysis data.

1.2 Decomposition of Concentration-Index

1.2.1 Indirect methods comparison (regression and predictive-forest SHAP)

From the indirect decomposition, we see that the largest regression contribution came from mother’s education: 23.1% under CI and 28.7% under CIg/CIc (Table 1). Father’s occupation in agriculture was second, contributing 17.9% under CI and 19.9% under CIg/CIc. In the predictive-forest SHAP decomposition, the contributions were less concentrated in one or two variables.

Residence and household wealth each contributed about 12% under CI, CIg and CIc, followed by mother’s occupation, father’s occupation, mother’s education and skilled birth attendance. The fitted regression model is reported in Table 3. Details of the fitted predictive forest are provided in Table 4 and Table 5, and its SHAP contributions are shown in Figure 10.

Table 1: Regression and predictive-forest SHAP percentage contributions by criterion.

Variable	Regression			Predictive-forest SHAP			
	CI (%)	CIg (%)	CIc (%)	CI (%)	CIg (%)	CIc (%)	L (%)
Mother’s education	23.058	28.726	28.726	10.421	10.421	10.421	11.678
Father’s occupation: Agriculture	17.863	19.885	19.885	9.753	9.753	9.753	10.364
Mother’s occupation: Agriculture	11.684	14.393	14.393	4.947	4.947	4.947	5.030
Household wealth	9.116	8.788	8.788	10.702	10.702	10.702	6.630
Type of residence	8.733	11.866	11.866	19.623	19.623	19.623	19.173
Skilled birth attendance	7.848	9.988	9.988	10.528	10.528	10.528	7.968
Region: Equateur	3.761	1.066	1.066	2.732	2.732	2.732	3.704
Father’s occupation: Household, unskilled, not working	3.370	0.955	0.955	-0.587	-0.587	-0.587	-0.465
Region: Kinshasa	3.221	0.647	0.647	2.340	2.340	2.340	3.811
Region: Sud-Kivu	2.006	0.213	0.213	-0.630	-0.630	-0.630	-0.522
Region: Kasai Oriental	1.746	0.513	0.513	0.169	0.169	0.169	0.110
Father’s education	1.462	0.869	0.869	2.837	2.837	2.837	3.129
Region: Maniema	1.075	0.083	0.083	2.433	2.433	2.433	3.387
Region: Orientale	0.984	0.251	0.251	0.038	0.038	0.038	-0.166
Region: Bas-Congo	0.678	0.060	0.060	0.170	0.170	0.170	0.346
Mother’s occupation: Household, unskilled, not working	0.599	0.306	0.306	0.050	0.050	0.050	-0.031
Birth order and interval: 5+ short interval	0.528	0.118	0.118	0.610	0.610	0.610	3.403
Region: Kasai Occidental	0.490	0.115	0.115	-0.037	-0.037	-0.037	-0.033
Birth order and interval: 5+ long interval	0.437	0.253	0.253	-0.477	-0.477	-0.477	-0.233
Sex of the child	0.418	0.454	0.454	0.126	0.126	0.126	0.025
Region: Katanga	0.296	0.067	0.067	-0.359	-0.359	-0.359	-2.043
Mother’s age at birth	0.261	0.249	0.249	0.530	0.530	0.530	0.717
Region: Nord-Kivu	0.208	0.018	0.018	0.307	0.307	0.307	0.157
Birth order and interval: 2-4 long interval	0.154	0.115	0.115	0.706	0.706	0.706	0.321
Birth order and interval: 2-4 short interval	0.004	0.001	0.001	0.331	0.331	0.331	0.177

1.3 Direct methods (concentration-index trees and forests)

1.3.1 Rank Dependent Concentration-Index Tree

The Corrected Concentration-Index Tree CI_c

We start with a tree based on the corrected concentration index, appropriate for bounded outcomes (Erreygers, 2009). In this method, the tree shows that inequality in under-five mortality is structured primarily by maternal education, father's agricultural occupation, household wealth, skilled birth attendance, and residence in Katanga Figure 2. The lowest-mortality terminal group consists of children whose mothers had some education and whose fathers were not in agriculture, with an under-five mortality rate of about 6.5%. At the opposite end, children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga had the highest mortality rate, about 17.8%. Thus, the corrected rank-dependent tree identifies a clear socioeconomic gradient, but one that is expressed through combinations of education, occupation, household wealth, delivery care, and region rather than through wealth alone.

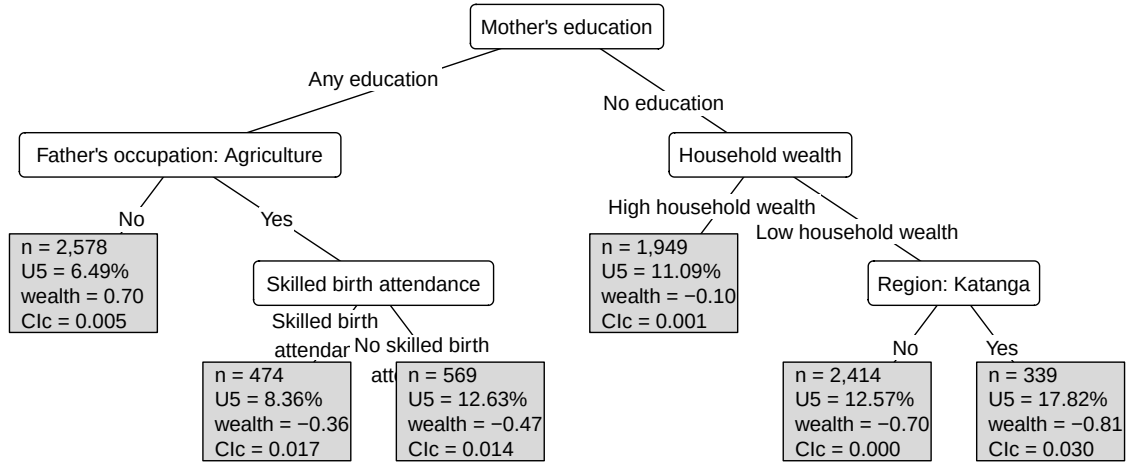


Figure 2: Concentration-index tree fitted with the CI_c criterion.

Generalised and standard concentration-index trees CI_g and CI

The appendix also shows the trees fitted using the generalised and standard concentration indices. The generalised concentration-index tree is almost identical to the corrected concentration-index tree Figure 13. It again separates children first by maternal education and father's agricultural occupation, then by skilled birth attendance, household wealth, and residence in Katanga. The mortality pattern is therefore substantively unchanged: the lowest-risk group consists of children whose mothers had some education and whose fathers were not in agriculture, while the highest-risk group consists of children whose mothers had no education, who lived in low-wealth households, and who resided in Katanga, with mortality around 17.8% Figure 13. The standard concentration-index tree gives a similar structure. However, it adds an extra distinction among children whose mothers had some education and whose fathers were not in agriculture, splitting them by short preceding birth interval Figure 14. Children in this otherwise advantaged group without a short birth interval have the lowest mortality, about 6.0%. In contrast, those with a short birth interval have higher mortality, about 11.7% Figure 14. Apart from this additional

demographic split, the standard CI tree points to the same main sources of inequality as the corrected and generalised trees: maternal education, father’s agricultural occupation, delivery care, household wealth, and residence in Katanga.

1.3.2 Level dependent Concentration-Index Tree L

We then fit a tree based on the level-dependent concentration index (Erreygers and Kessels, 2017) criterion, L . In this method, the selected tree shows that inequality in under-five mortality is structured primarily by rural residence, father’s agricultural occupation, household wealth, and province, especially Kinshasa, Equateur, and Katanga Figure 3. The lowest-mortality terminal group consists of urban children in Kinshasa whose fathers were not in agriculture, with an under-five mortality rate of about 5.1%. The highest-mortality terminal group consists of rural children living in low-wealth households in Katanga, with an under-five mortality rate of about 18.5%. Thus, the level-dependent tree tells a more spatial story of inequality: mortality risk is first organised by the rural-urban divide, then by occupation, wealth, and region. This shows that the level-dependent method found a different structure compared to the rank-dependent methods

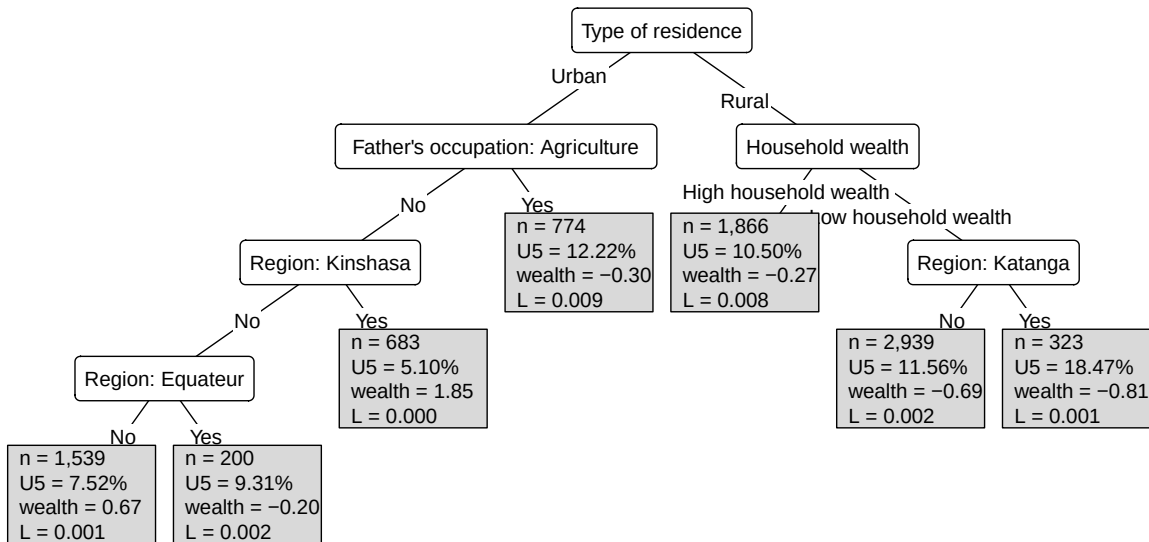


Figure 3: Level dependent concentration-index tree for under-five mortality in the DRC.

Tree validation plots

The rank-dependent trees performed well on the training data but poorly on the validation data. The CI, CIc, and CIg trees removed approximately 87 to 88% of root-node inequality during training, but had negative validation gains at the selected settings: -75.5% for CI and -73.1% for both CIc and CIg Figure 4, Table 7. In this criterion, a negative validation gain means that the sum of the terminal-node concentration-index values in the validation fold exceeded the concentration-index value at the root node. Thus, although these trees identified substantial rank-based structure in the fitted sample, the same partitions did not reduce inequality when applied to held-out folds. Tree complexity decreased as the minimum relative split gain increased, since larger thresholds allowed only splits that removed a larger share of parent-node impurity

Figure 12. The selected trees remained small, with an average of 7.5 terminal nodes for CI, 7.3 for CIc and CIg, and 6.8 for L Table 8, Figure 12. Unlike rank-dependent trees, the level-dependent tree maintained a positive validation gain throughout the tuning path. At the selected setting, it removed 70.5% of root-node inequality in held-out folds while retaining an average of 6.8 terminal nodes Figure 4, Table 7. The rural-urban, occupational, wealth, and regional partition identified by the L tree was therefore more stable under the validation criterion used in this thesis.

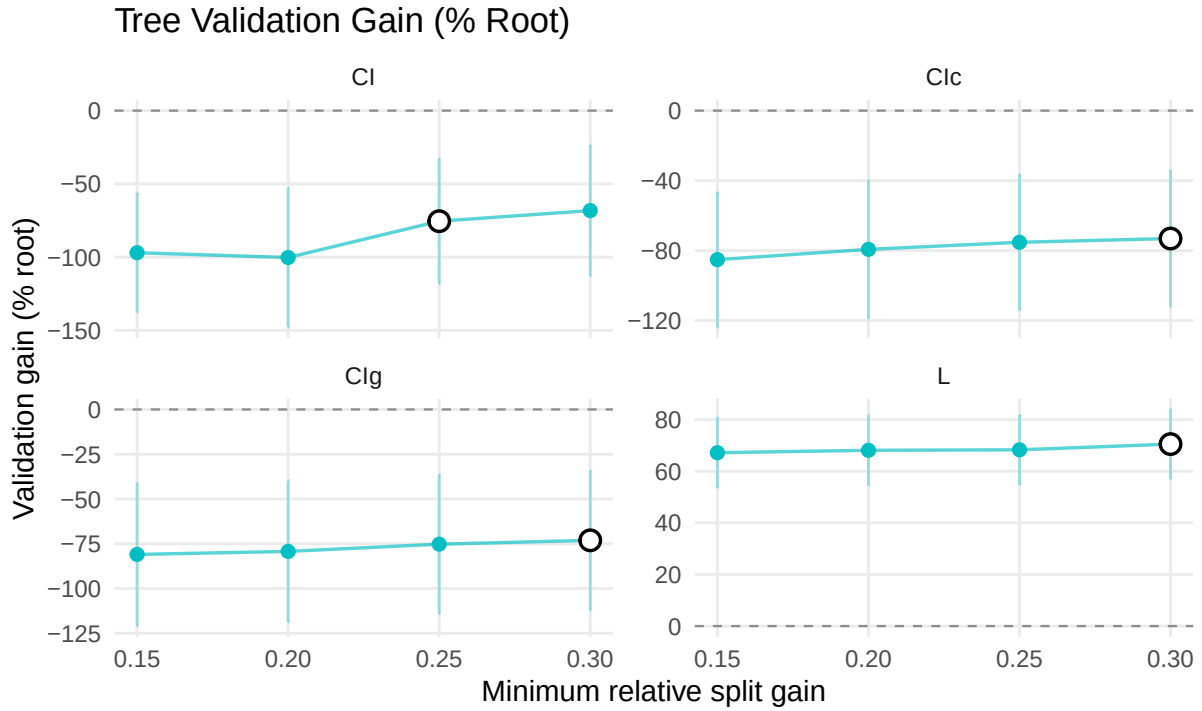


Figure 4: Tree validation gain and complexity by minimum relative split gain.

1.3.3 Surrogate Tree Comparison

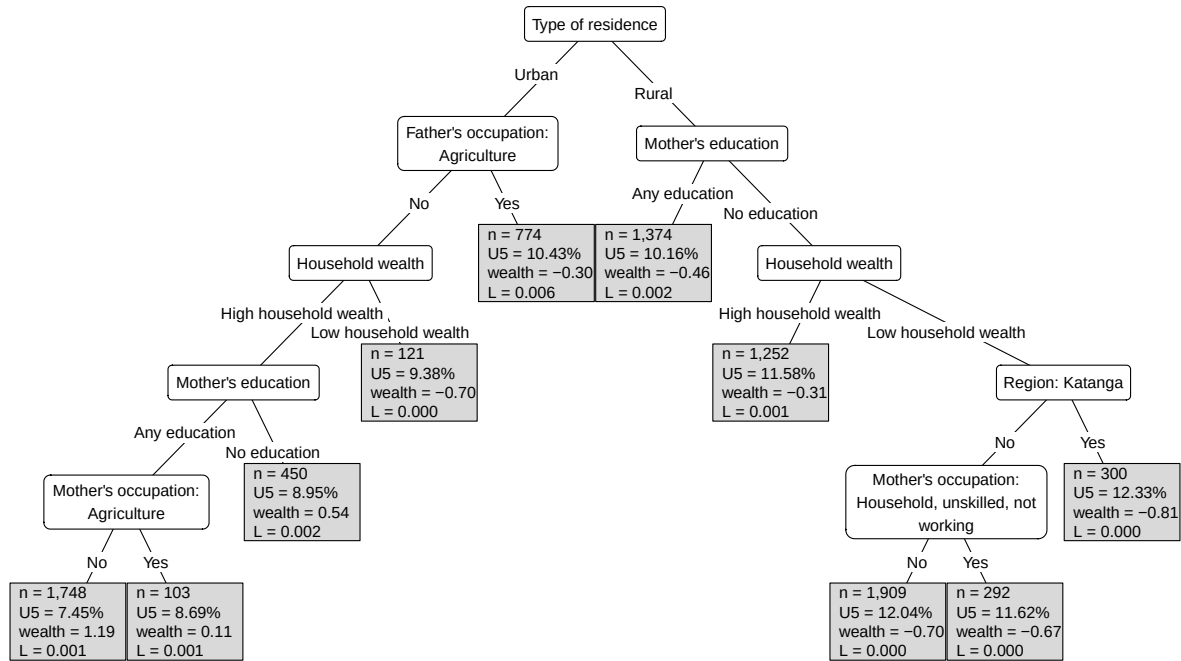


Figure 5: Surrogate tree summarising the selected concentration-index forest predictions.

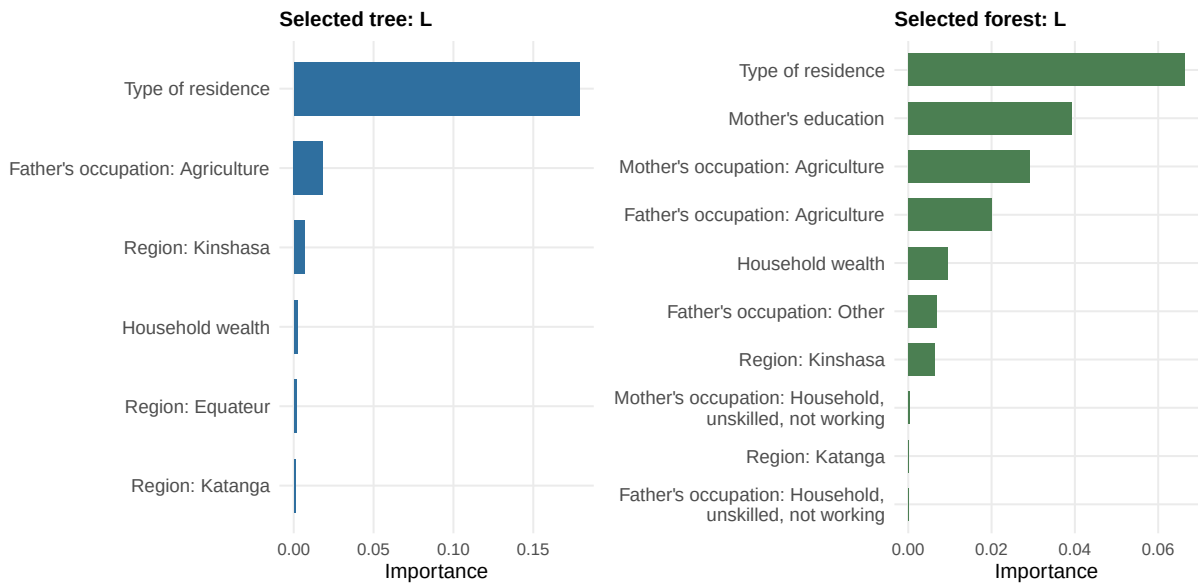


Figure 6: Variable importance for the selected concentration-index tree and forest.

1.4 Simulation Study

1.4.1 Recovery of the Simulated Subgroup Structure

The simulation study is used as a direct check on the tree-growing rule. In the null simulation, wealth, the binary outcome, and the candidate subgroup variables are generated independently.

In the positive-control simulation, the outcome is concentrated among poorer observations in a planted subgroup defined by residence, maternal education, and province. The null trees should therefore remain at the root node, while the positive-control tree should recover a readable subgroup structure.

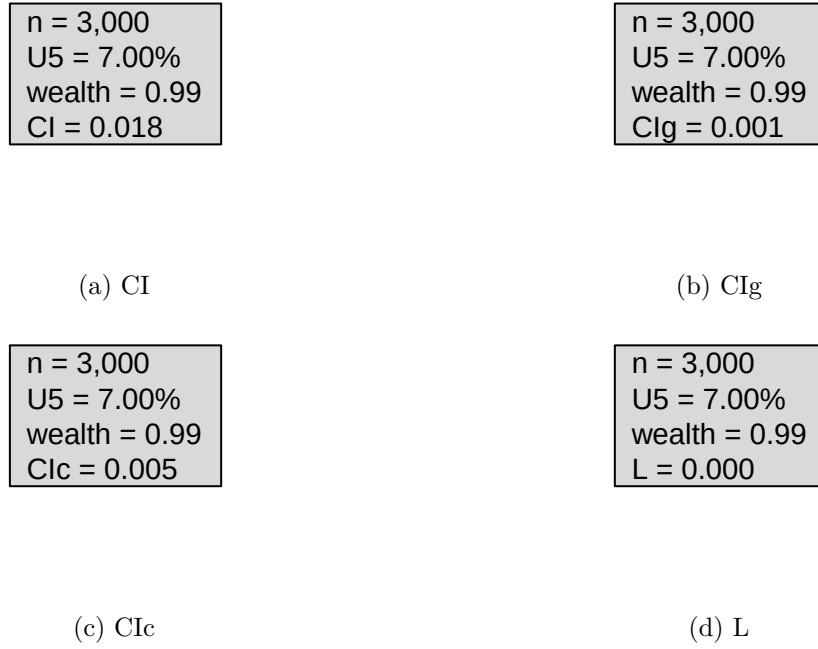


Figure 7: Null simulation concentration-index trees by criterion.

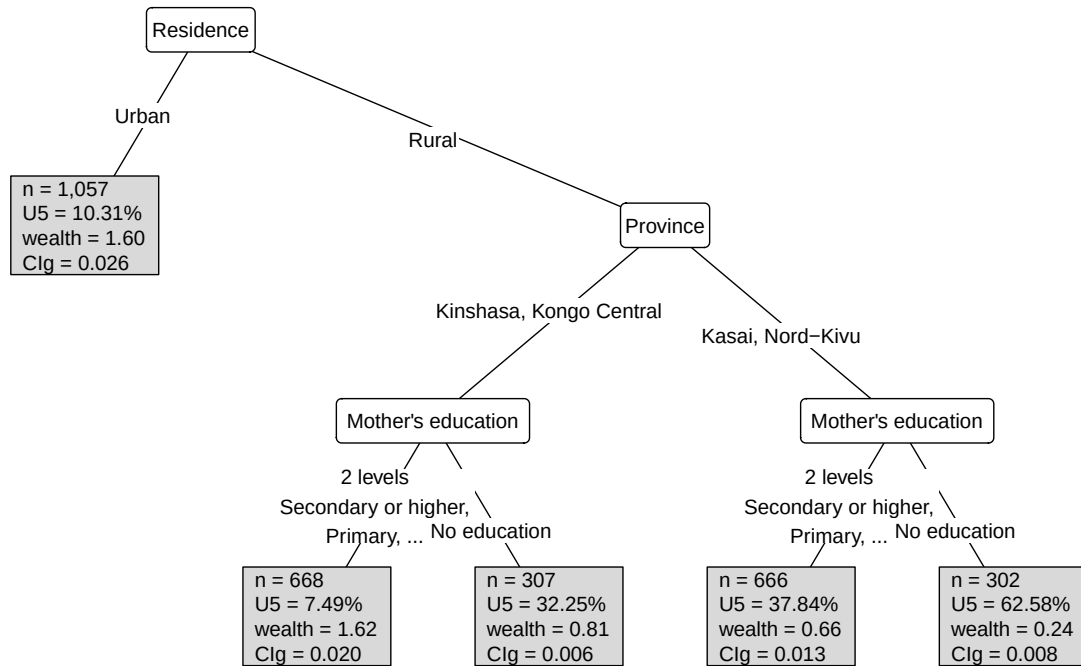


Figure 8: CIg concentration-index tree recovered from simulated data with a known subgroup-generating mechanism.

2 Appendix

2.1 Results

2.1.1 Appendix R1: Exploratory Data Outputs

Table 2: Weighted distribution of analysis variables by under-five mortality status

Variable	Under-five mortality		Overall
	Alive at age 5	Died before age 5	Overall
	(weighted N=7,464)	(weighted N=859)	(weighted N=8,323)
Household wealth			
High household wealth	4,266.9 (57.2%)	418.6 (48.7%)	4,685.5 (56.3%)
Low household wealth	3,197.4 (42.8%)	440.2 (51.3%)	3,637.6 (43.7%)
Skilled.birth.attendance			
Skilled.birth.attendance.1	3,225.3 (43.2%)	295.1 (34.4%)	3,520.5 (42.3%)
No.skilled.birth.attendance	4,239 (56.8%)	563.6 (65.6%)	4,802.6 (57.7%)
Sex of the child			

Female	3,828.8 (51.3%)	390.4 (45.5%)	4,219.2 (50.7%)
Male	3,635.5 (48.7%)	468.4 (54.5%)	4,103.9 (49.3%)
Birth order and interval			
First birth	1,396.8 (18.7%)	190.9 (22.2%)	1,587.7 (19.1%)
2-4 short interval	774.3 (10.4%)	110.1 (12.8%)	884.4 (10.6%)
2-4 long interval	2,603.8 (34.9%)	216.6 (25.2%)	2,820.3 (33.9%)
5+ short interval	696.6 (9.3%)	150.6 (17.5%)	847.2 (10.2%)
5+ long interval	1,992.8 (26.7%)	190.7 (22.2%)	2,183.5 (26.2%)
Mother's age at birth			
20 or more	4,256.1 (57.0%)	472.6 (55.0%)	4,728.8 (56.8%)
Less than 20	3,208.1 (43.0%)	386.2 (45.0%)	3,594.3 (43.2%)
Type of residence			
Urban	2,932.1 (39.3%)	263.6 (30.7%)	3,195.7 (38.4%)
Rural	4,532.2 (60.7%)	595.2 (69.3%)	5,127.4 (61.6%)
Mother's education			
Any education	3,342.9 (44.8%)	278.8 (32.5%)	3,621.8 (43.5%)
No education	4,121.4 (55.2%)	579.9 (67.5%)	4,701.3 (56.5%)
Father's education			
Any education	5,490 (73.6%)	589.1 (68.6%)	6,079.2 (73.0%)
No education	1,974.3 (26.4%)	269.7 (31.4%)	2,243.9 (27.0%)
Mother's occupation			
Other	1,610.2 (21.6%)	135.1 (15.7%)	1,745.4 (21.0%)
Household, unskilled, not working	1,757.2 (23.5%)	172.1 (20.0%)	1,929.4 (23.2%)
Agriculture	4,096.8 (54.9%)	551.6 (64.2%)	4,648.4 (55.8%)
Father's occupation			
Other	2,810.3 (37.6%)	242.1 (28.2%)	3,052.4 (36.7%)
Household, unskilled, not working	961.3 (12.9%)	108.5 (12.6%)	1,069.8 (12.9%)
Agriculture	3,692.7 (49.5%)	508.1 (59.2%)	4,200.9 (50.5%)

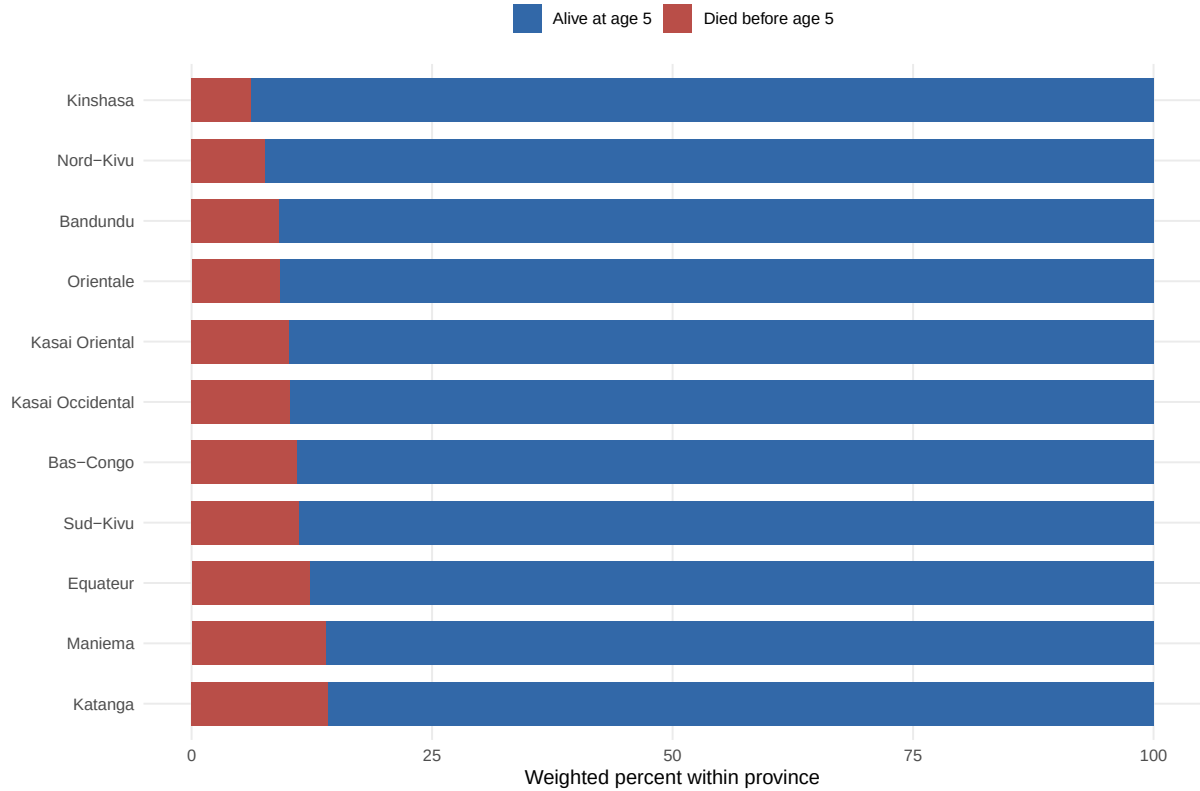


Figure 9: Weighted province distribution by under-five mortality status.

2.1.2 Appendix R2: Regression and Predictive-Forest Details

Regression model summary

Table 3: Survey-weighted regression model summary

Term	Estimate	Std. error	Statistic	p-value	Conf. low	Conf. high
Intercept	-2.908	0.275	-10.563	<0.001	-3.448	-2.369
Household wealth	0.068	0.125	0.544	0.5865	-0.177	0.314
Skilled birth attendance	0.198	0.111	1.783	0.0746	-0.020	0.416
Sex of the child	0.242	0.105	2.294	0.0218	0.035	0.449
Birth order and interval: 2-4 short interval	-0.004	0.186	-0.020	0.9838	-0.367	0.360
Birth order and interval: 2-4 long interval	-0.509	0.169	-3.013	0.0026	-0.840	-0.178
Birth order and interval: 5+ short interval	0.420	0.210	1.995	0.0461	0.007	0.832
Birth order and interval: 5+ long interval	-0.396	0.211	-1.882	0.0599	-0.809	0.017
Mother's age at birth	0.033	0.143	0.229	0.8186	-0.247	0.312
Type of residence	0.090	0.154	0.584	0.5591	-0.212	0.392
Mother's education	0.336	0.132	2.555	0.0106	0.078	0.594

Father's education	-0.036	0.128	-0.280	0.7798	-0.286	0.214
Mother's occupation: Household, unskilled, not working	-0.018	0.155	-0.114	0.9093	-0.322	0.287
Mother's occupation: Agriculture	0.140	0.163	0.861	0.3895	-0.180	0.460
Father's occupation: Household, unskilled, not working	0.235	0.162	1.450	0.1472	-0.083	0.554
Father's occupation: Agriculture	0.225	0.149	1.509	0.1314	-0.067	0.517
Region: Bas-Congo	0.258	0.243	1.062	0.2883	-0.218	0.734
Region: Equateur	0.250	0.238	1.051	0.2934	-0.216	0.715
Region: Kasai Occidental	0.105	0.228	0.460	0.6452	-0.342	0.553
Region: Kasai Oriental	0.153	0.240	0.635	0.5251	-0.318	0.624
Region: Katanga	0.515	0.228	2.258	0.0240	0.068	0.962
Region: Kinshasa	0.081	0.262	0.310	0.7564	-0.433	0.596
Region: Maniema	0.489	0.224	2.178	0.0294	0.049	0.929
Region: Nord-Kivu	-0.252	0.294	-0.857	0.3915	-0.827	0.324
Region: Orientale	-0.084	0.265	-0.316	0.7522	-0.603	0.435
Region: Sud-Kivu	0.300	0.270	1.111	0.2667	-0.229	0.829

Predictive ranger tuning results

Table 4: Weighted class distribution before and after ranger undersampling

Sample	Outcome	Rows	Weighted N	Weighted percent
Before undersampling	Alive at age 5	7424	7464.30	89.68
Before undersampling	Died before age 5	872	858.80	10.32
After undersampling	Alive at age 5	1274	1289.74	60.03
After undersampling	Died before age 5	872	858.80	39.97

Table 5: Cross-validation results for the predictive ranger model

Metric	mtry	min_n	Mean	Std. error	Folds	Rank
accuracy	22	1100	0.6132	0.0035	10	1
accuracy	29	1100	0.6123	0.0042	10	2
accuracy	10	200	0.6108	0.0090	10	3
accuracy	16	1100	0.6100	0.0030	10	4
accuracy	29	200	0.6090	0.0106	10	5
accuracy	22	200	0.6080	0.0098	10	6
accuracy	10	1100	0.6072	0.0023	10	7
accuracy	16	200	0.6066	0.0084	10	8

accuracy	10	2000	0.5937	0.0003	10	9
accuracy	16	2000	0.5937	0.0003	10	10
accuracy	22	2000	0.5937	0.0003	10	11
accuracy	29	2000	0.5937	0.0003	10	12
roc_auc	10	200	0.6155	0.0111	10	1
roc_auc	16	200	0.6140	0.0104	10	2
roc_auc	22	200	0.6132	0.0105	10	3
roc_auc	29	200	0.6126	0.0103	10	4
roc_auc	10	1100	0.6079	0.0108	10	5
roc_auc	29	1100	0.6053	0.0108	10	6
roc_auc	22	1100	0.6052	0.0108	10	7
roc_auc	16	1100	0.6048	0.0110	10	8
roc_auc	10	2000	0.5000	0.0000	10	9
roc_auc	16	2000	0.5000	0.0000	10	10
roc_auc	22	2000	0.5000	0.0000	10	11
roc_auc	29	2000	0.5000	0.0000	10	12
sensitivity	29	200	0.3072	0.0162	10	1
sensitivity	22	200	0.2923	0.0188	10	2
sensitivity	16	200	0.2900	0.0143	10	3
sensitivity	10	200	0.2774	0.0152	10	4
sensitivity	29	1100	0.1100	0.0105	10	5
sensitivity	22	1100	0.1019	0.0138	10	6
sensitivity	16	1100	0.0859	0.0120	10	7
sensitivity	10	1100	0.0597	0.0089	10	8
sensitivity	10	2000	0.0000	0.0000	10	9
sensitivity	16	2000	0.0000	0.0000	10	10
sensitivity	22	2000	0.0000	0.0000	10	11
sensitivity	29	2000	0.0000	0.0000	10	12

Predictive ranger SHAP summary plot

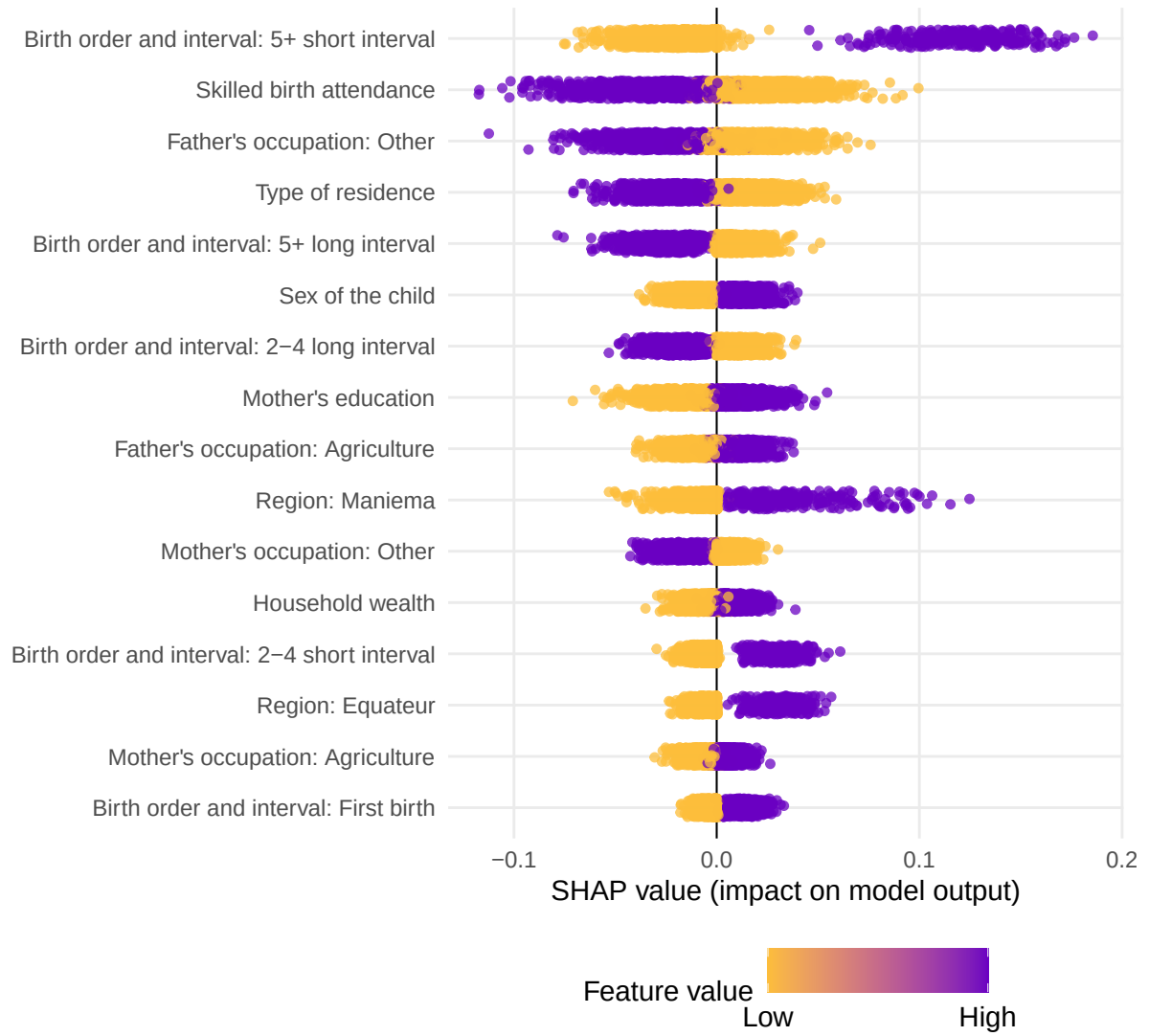


Figure 10: SHAP summary plot for the predictive ranger model.

2.1.3 Appendix R3: Concentration-Index Tree Details

Tree tuning and selection details

Table 6: Best tree hyperparameters selected by validation gain within each impurity criterion

type	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	500	100	0.01	10	1e-04	0.25
CIg	500	100	0.01	10	1e-04	0.30
CIc	500	100	0.01	10	1e-04	0.30
L	500	100	0.01	10	1e-04	0.30

Tree fit and selection summaries

Table 7: Tree summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1191	NA
CI	Training gain	0.1013	0.0017
CI	Training gain (% root)	87.9025	1.5310
CI	Validation gain	-0.0235	0.0220
CI	Validation gain (% root)	-75.4763	42.7213
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0417	0.0008
CIc	Training gain (% root)	87.5995	1.4760
CIc	Validation gain	-0.0107	0.0134
CIc	Validation gain (% root)	-73.1067	39.0032
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0104	0.0002
CIg	Training gain (% root)	87.5995	1.4760
CIg	Validation gain	-0.0027	0.0034
CIg	Validation gain (% root)	-73.1067	39.0032
L	Root impurity	0.3128	NA
L	Training gain	0.1906	0.0061
L	Training gain (% root)	98.1536	0.2271
L	Validation gain	0.2861	0.1077
L	Validation gain (% root)	70.5254	13.4254

Tree selection metrics

Table 8: Selection metrics for fitted concentration-index trees

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Terminal nodes
CI	0.1191	0.1013	0.8790	-0.0235	-0.7548	7.5
CIc	0.0483	0.0417	0.8760	-0.0107	-0.7311	7.3
CIg	0.0121	0.0104	0.8760	-0.0027	-0.7311	7.3
L	0.3128	0.1906	0.9815	0.2861	0.7053	6.8

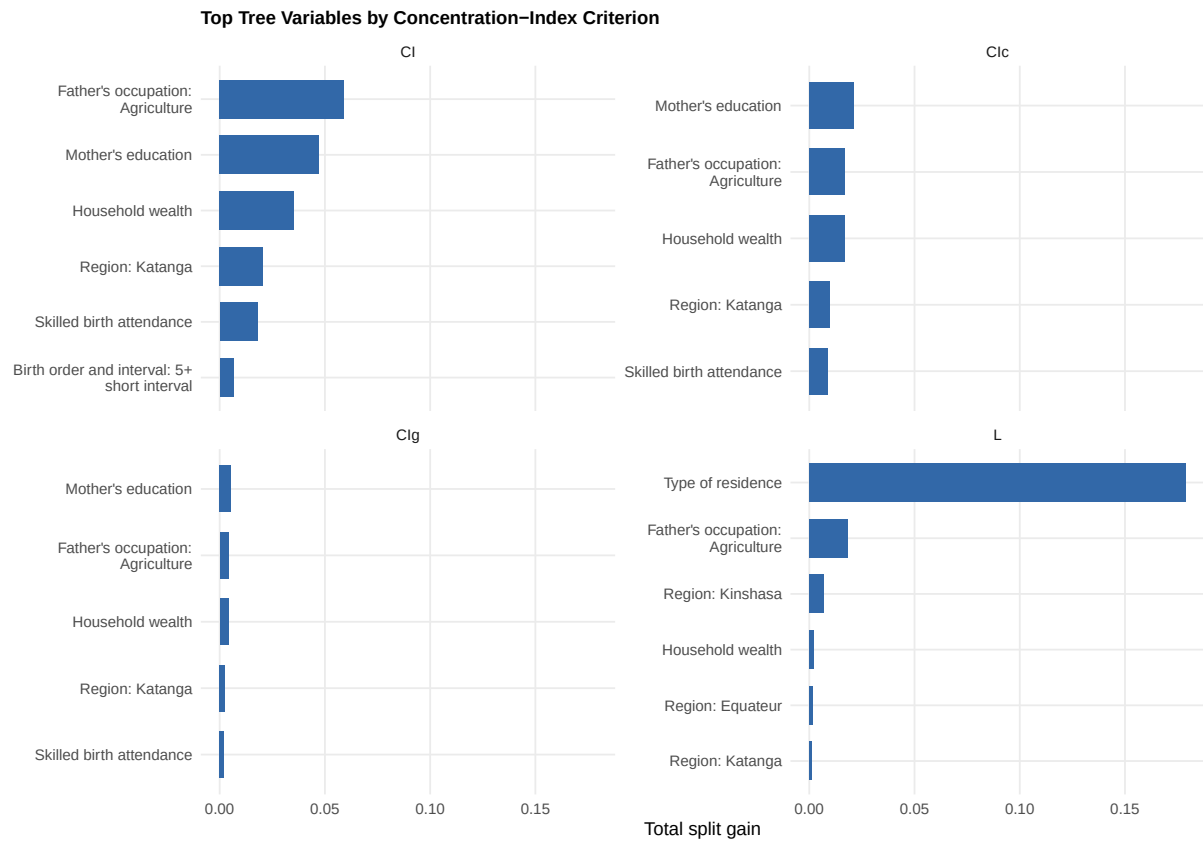


Figure 11: Top tree variables by concentration-index criterion.

Tree complexity plots

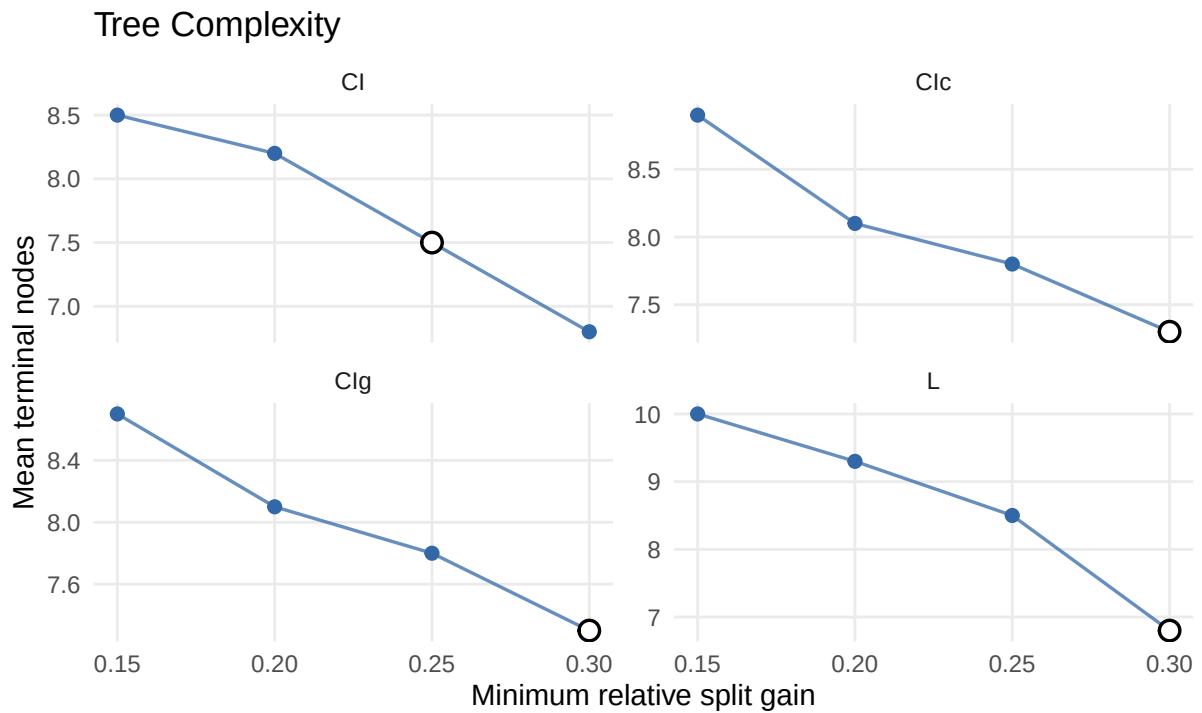


Figure 12: Tree complexity by number of terminal nodes.

2.1.4 Additional Concentration-Index Tree Plots

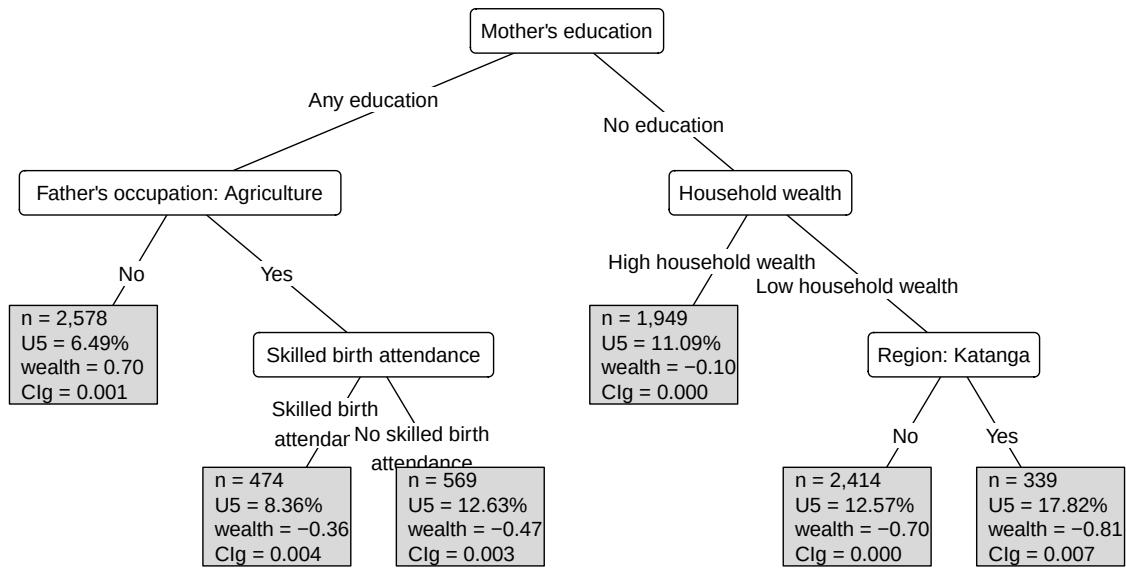


Figure 13: Concentration-index tree fitted with the CIg criterion.

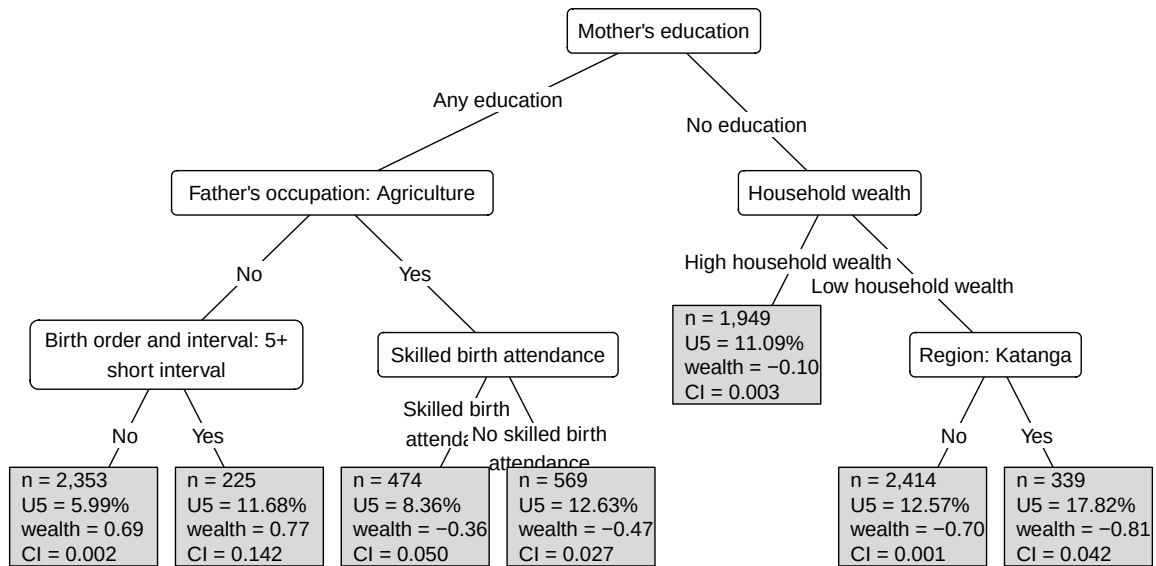


Figure 14: Concentration-index tree fitted with the CI criterion.

2.1.5 Appendix R4: Rpart Comparison Tree Details

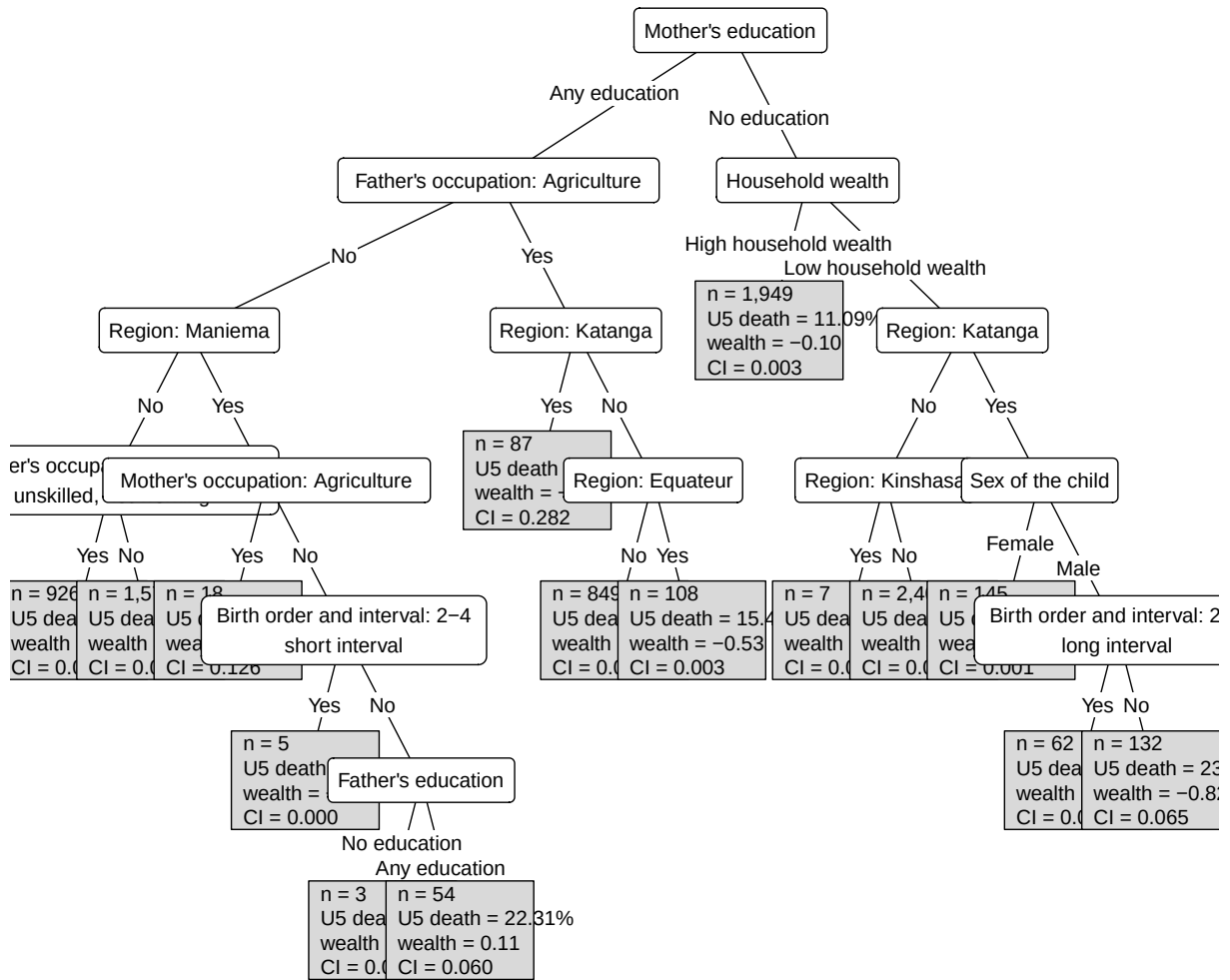


Figure 15: Lecturer rpart comparison tree using absolute CI impurity.

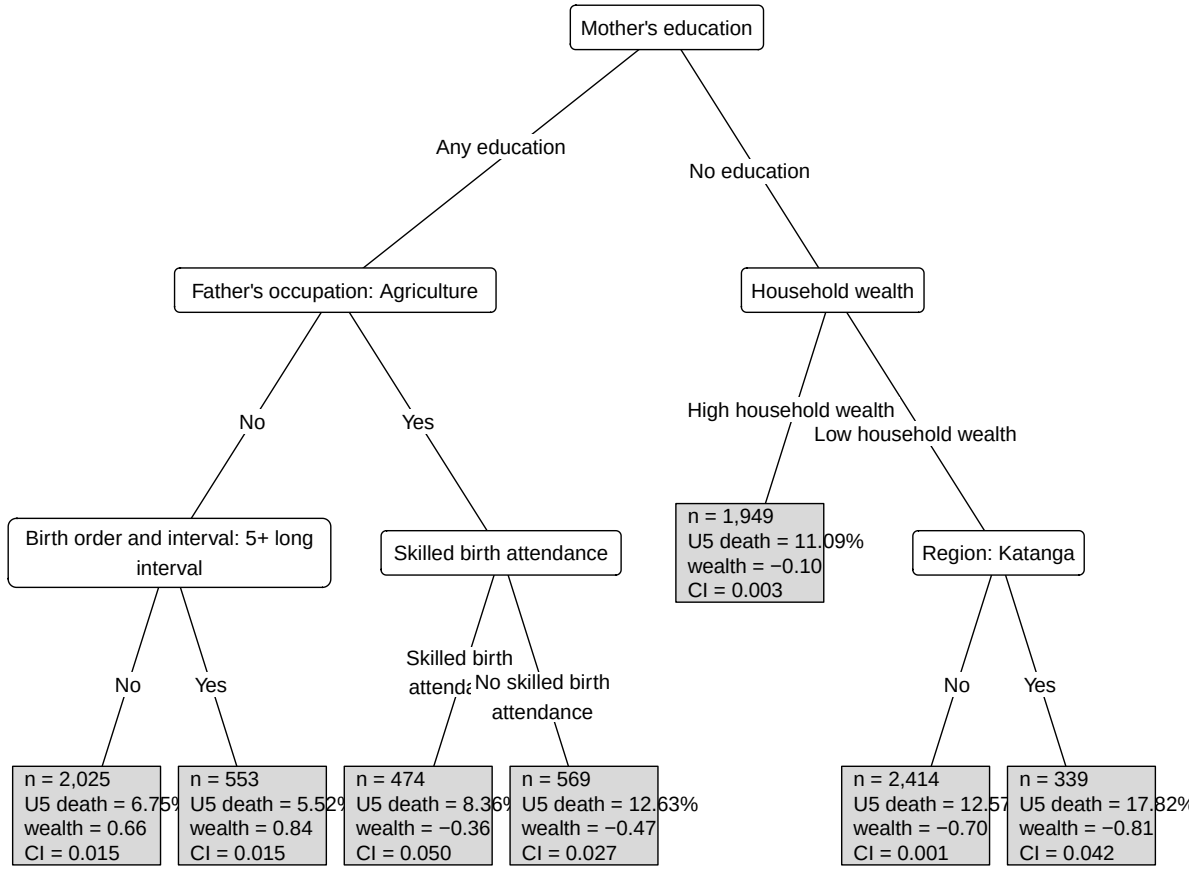


Figure 16: Lecturer rpart comparison tree using robust absolute CI impurity.

2.1.6 Appendix 5: Concentration-Index Forest Details

Forest tuning and selection details

Table 9: Best forest hyperparameters selected by validation gain within each impurity criterion

type	ntree	mtry	minsplit	minbucket	minprob	maxdepth	min_gain	min_relative_gain
CI	50	10	500	100	0.01	15	0	0.3
CIc	50	10	500	100	0.01	15	0	0.3
CIg	50	10	500	100	0.01	15	0	0.2
L	50	10	500	100	0.01	15	0	0.4

Forest fit and selection summaries

Table 10: Forest summary for the selected settings

criterion	metric	estimate	std_error
CI	Root impurity	0.1191	NA
CI	Training gain	0.0222	0.0027
CI	Training gain (% root)	19.4745	2.4825
CI	Validation gain	-0.0032	0.0077
CI	Validation gain (% root)	-13.8147	8.1216
CIc	Root impurity	0.0483	NA
CIc	Training gain	0.0167	0.0007
CIc	Training gain (% root)	35.0907	1.5278
CIc	Validation gain	-0.0041	0.0052
CIc	Validation gain (% root)	-37.0209	19.7261
CIg	Root impurity	0.0121	NA
CIg	Training gain	0.0047	0.0002
CIg	Training gain (% root)	39.8170	1.2803
CIg	Validation gain	-0.0010	0.0014
CIg	Validation gain (% root)	-37.6958	20.8545
L	Root impurity	0.3128	NA
L	Training gain	0.1803	0.0059
L	Training gain (% root)	92.8313	0.4426
L	Validation gain	0.2803	0.1025
L	Validation gain (% root)	71.8405	13.3582

Forest selection metrics

Table 11: Selection metrics for fitted concentration-index forests

Criterion	Root objective	Train gain	Train gain/root	Validation gain	Validation gain/root	Mean terminal nodes
CI	0.1191	0.0222	0.1947	-0.0032	-0.1381	1.846
CIc	0.0483	0.0167	0.3509	-0.0041	-0.3702	2.700
CIg	0.0121	0.0047	0.3982	-0.0010	-0.3770	3.196
L	0.3128	0.1803	0.9283	0.2803	0.7184	3.270

Forest validation plots

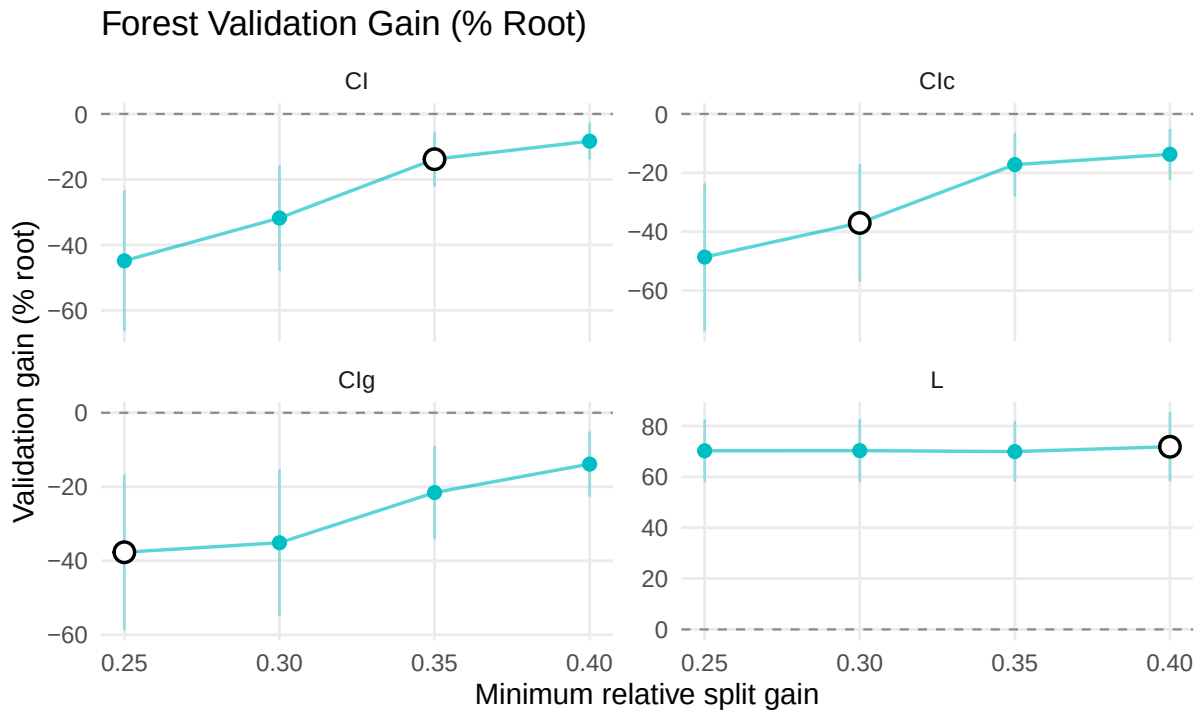


Figure 17: Forest validation gain and complexity by minimum relative split gain.

Forest complexity plots

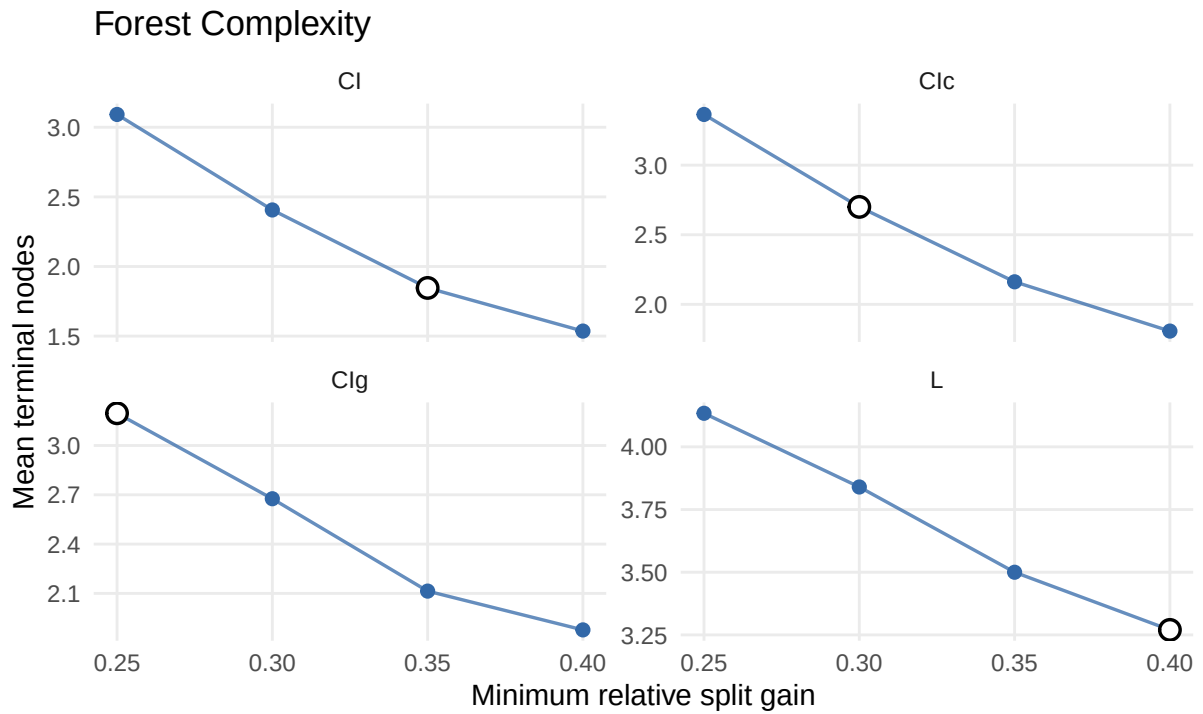


Figure 18: Forest complexity by number of terminal nodes.

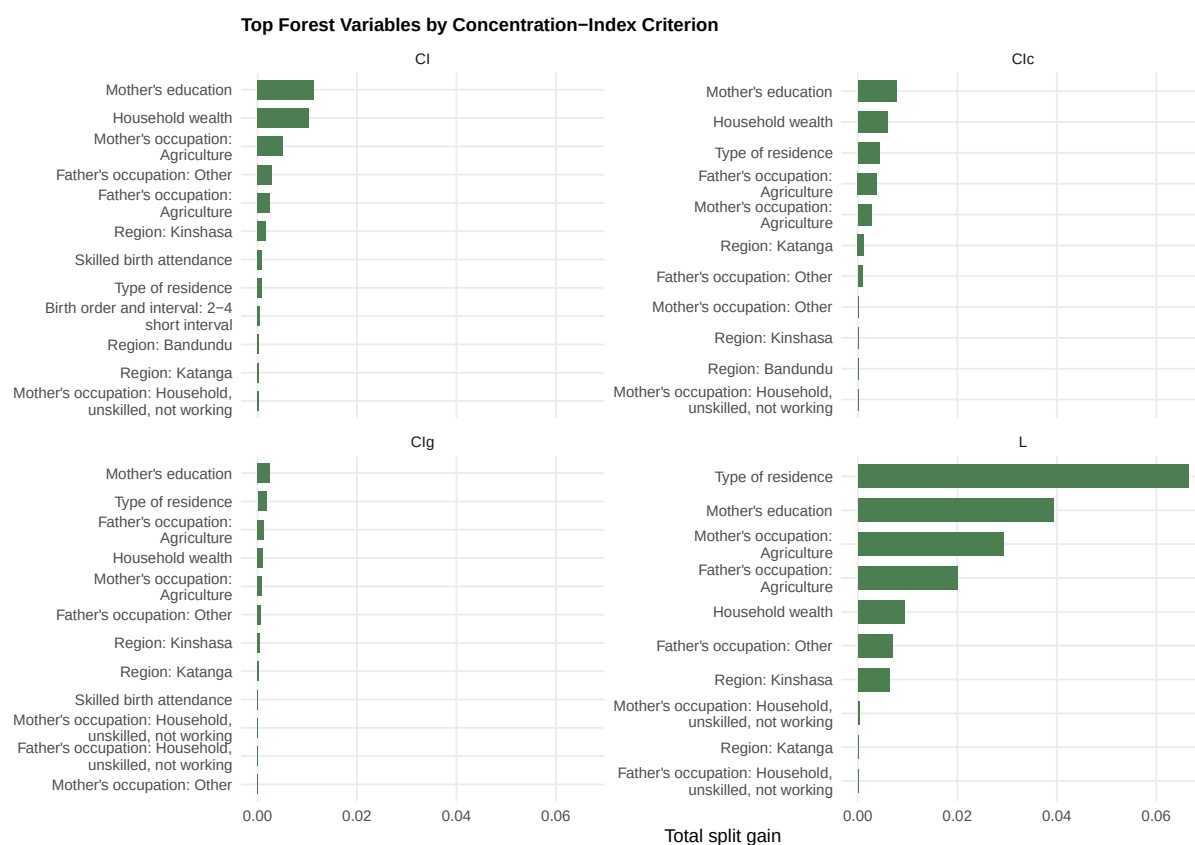


Figure 19: Top forest variables by concentration-index criterion.

Forest Surrogate Trees plots

The selected forest surrogate appears in the results section. The remaining criterion-specific forest surrogates are retained here for comparison.

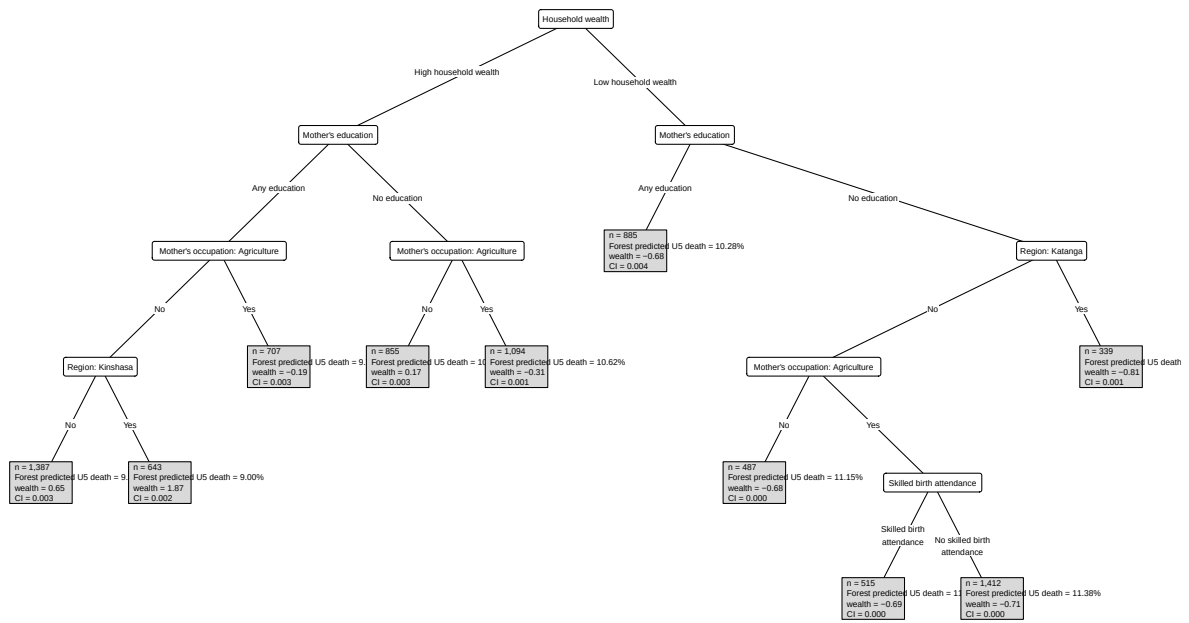


Figure 20: Surrogate tree for the CI concentration-index forest.

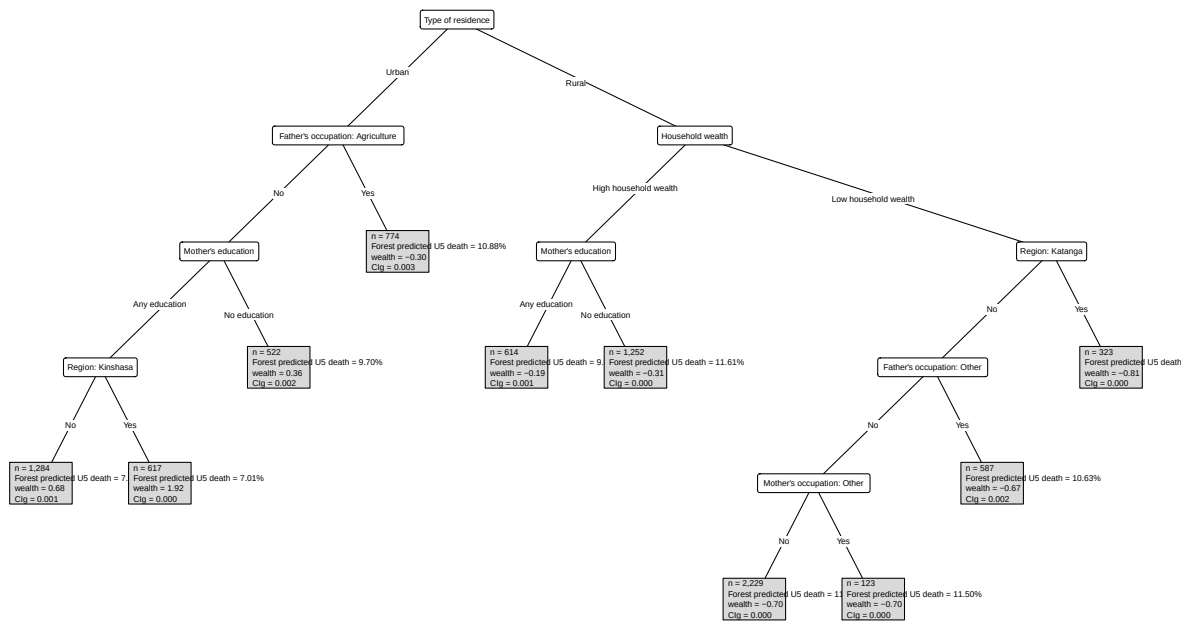


Figure 21: Surrogate tree for the Clg concentration-index forest.

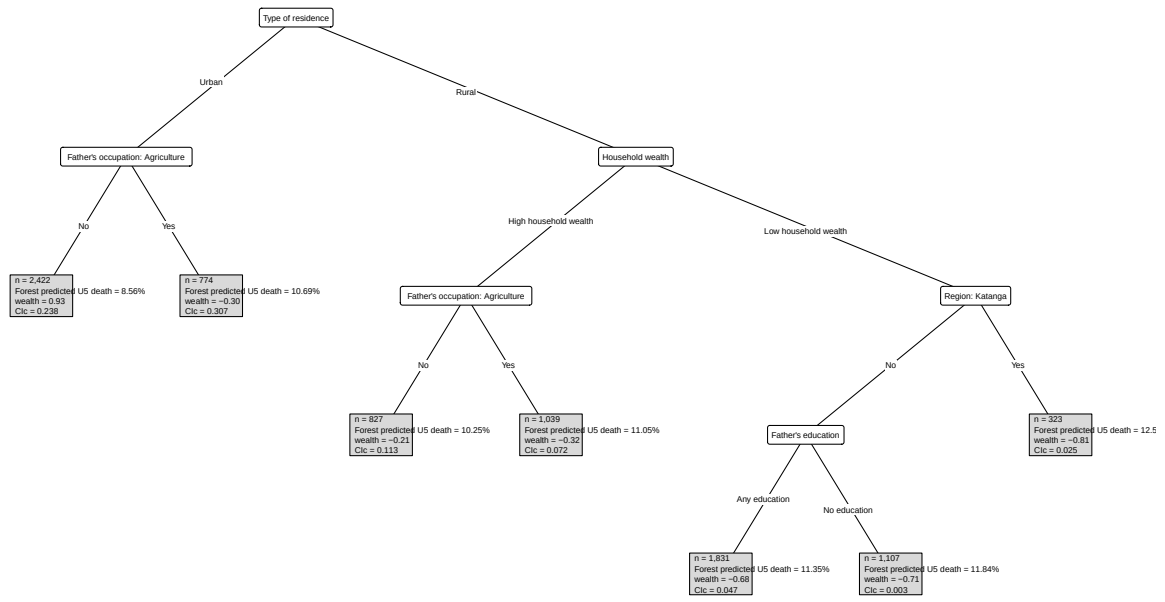


Figure 22: Surrogate tree for the CIc concentration-index forest.

2.1.7 Appendix S: Simulation Sensitivity Plots

The CIg tree is shown in the main results section. The remaining positive-control simulation trees are shown here to check whether the recovered subgroup pattern is stable across concentration-index criteria.

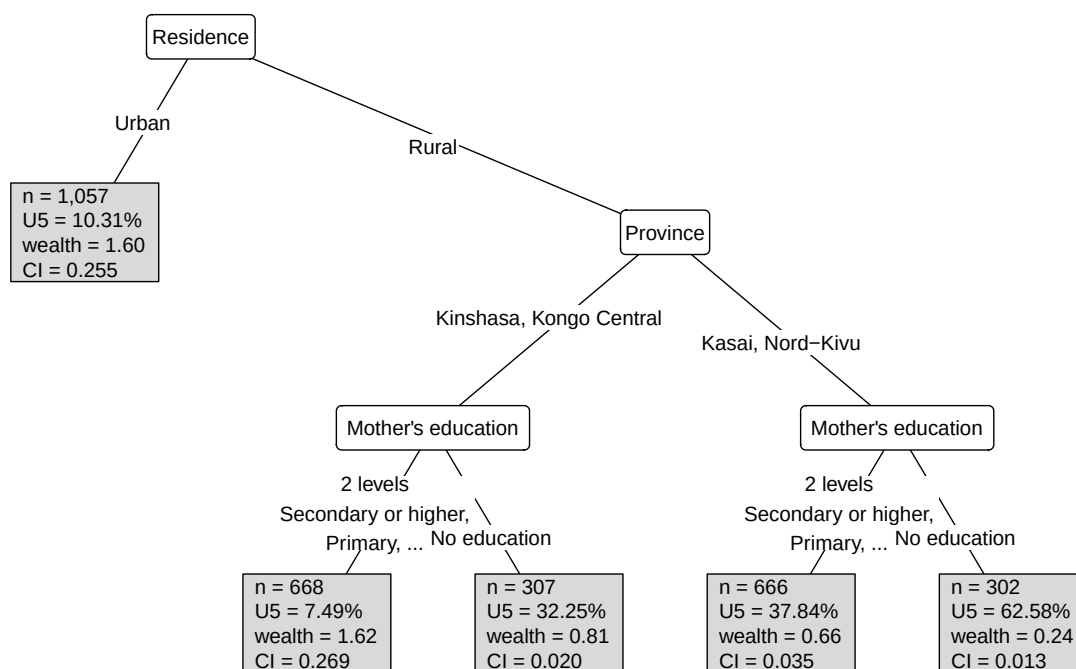


Figure 23: Positive-control simulation tree fitted with the CI criterion.

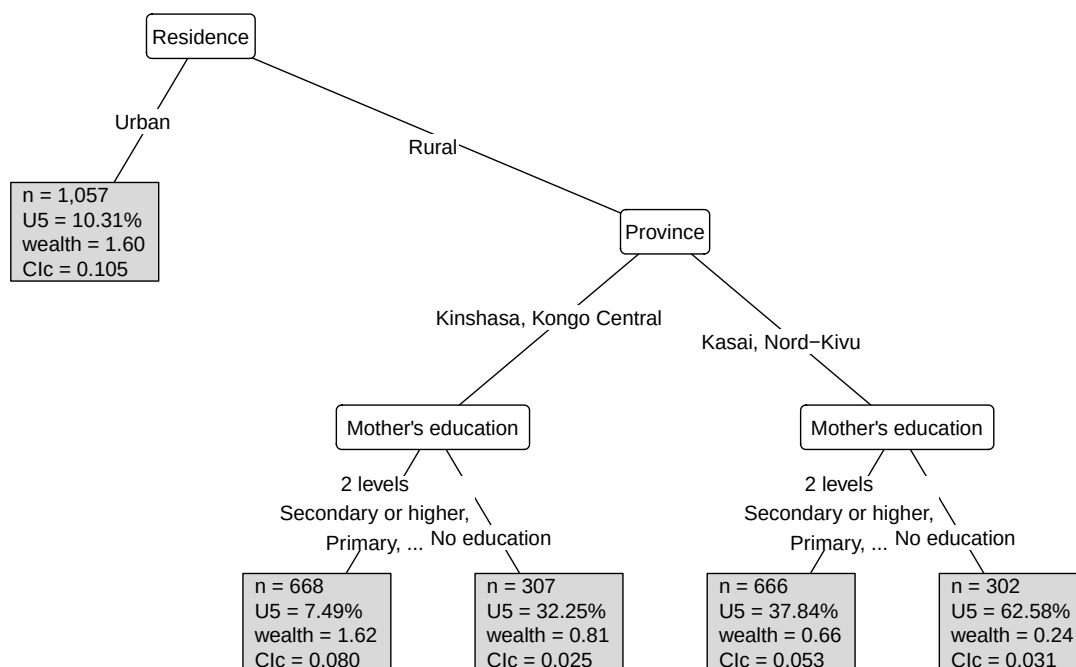


Figure 24: Positive-control simulation tree fitted with the Clc criterion.

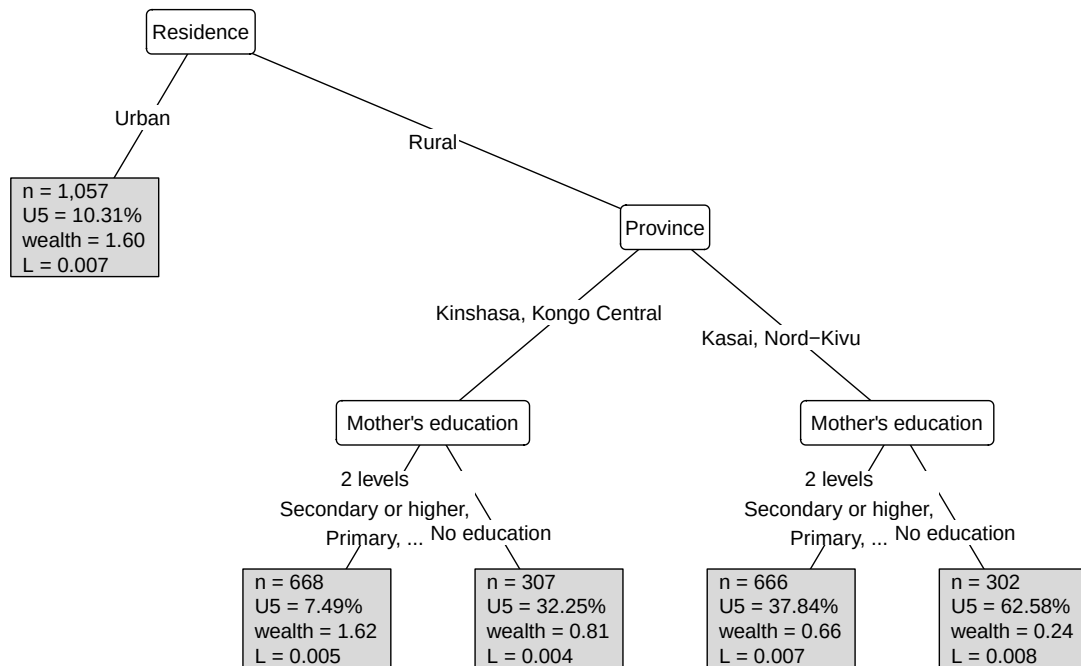


Figure 25: Positive-control simulation tree fitted with the L criterion.

Erreygers, G. (2009) ‘Correcting the concentration index’, *Journal of Health Economics*, 28(2), pp. 504–515. Available at: <https://doi.org/10.1016/j.jhealeco.2008.02.003>.

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